

NOGALES, AZ, USA

WMO: 722728

Lat: 31.421N

Lon: 110.846W

Elev: 1199

StdP: 87.73

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03196

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.1	-1.5	-19.7	0.8	14.5	-17.3	0.9	14.5	10.9	13.0	9.3	13.7	1.2	70	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	17.8	37.8	17.0	36.6	17.0	35.2	16.8	21.4	28.9	20.8	28.2	20.4	27.7	4.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.8	16.8	22.7	19.1	16.1	22.6	18.6	15.6	22.5	68.0	29.0	66.1	28.1	64.3	27.5	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.2	-6.9	40.5	2.1	1.5	-8.4	41.6	-9.6	42.5	-10.8	43.3	-12.4	44.4		
(5)			-8.6	22.5	2.0	0.7	-10.0	23.1	-11.2	23.5	-12.3	23.9	-13.8	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	9.9	10.9	13.9	16.9	20.8	26.5	26.9	25.7	23.9	19.0	14.0	9.6		
	DBStd	6.97	3.31	3.64	3.47	3.23	3.13	2.54	2.11	1.98	2.02	2.97	3.52	3.51		
	HDD10.0	143	42	30	9	1	0	0	0	0	0	1	9	51		
	HDD18.3	1117	262	207	139	63	14	0	0	0	0	27	133	271		
	CDD10.0	3136	38	56	132	207	333	496	524	488	416	281	128	38		
	CDD18.3	1069	0	0	3	19	89	246	266	229	166	49	2	0		
	CDH23.3	14055	9	56	231	619	1654	3657	2786	2216	1776	863	173	16		
(13)	CDH26.7	7024	0	6	41	180	772	2225	1510	1119	842	304	24	0		
(14)	Wind	WSAvg	2.5	2.2	2.6	2.8	3.1	3.2	3.0	2.3	2.0	2.2	2.4	2.3		
(15) Precipitation	PrecAvg	433	27	22	24	5	4	10	118	89	48	32	18	41		
	PrecMax	682	80	64	97	23	23	56	234	200	132	211	88	203		
	PrecMin	252	0	0	0	0	0	0	18	31	0	0	0	0		
	PrecStd	114	23	21	25	6	7	14	51	46	32	48	21	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	27.5	29.9	32.4	36.7	40.2	39.1	37.5	35.6	33.1	29.1	25.2		
		MCWB	9.7	10.4	11.6	12.8	15.3	17.0	17.8	19.0	17.6	15.1	12.0	10.0		
	2%	DB	22.7	25.1	27.7	30.4	34.2	38.2	37.4	35.9	34.1	31.5	27.2	22.9		
		MCWB	8.8	9.7	11.2	11.9	13.9	16.4	17.9	19.1	17.2	14.8	11.6	9.3		
	5%	DB	21.1	22.8	26.0	28.8	32.7	37.1	35.9	34.2	32.9	29.9	25.5	20.9		
		MCWB	8.2	8.9	10.3	11.4	13.4	16.1	18.0	19.2	17.4	14.3	11.2	8.8		
	10%	DB	18.9	20.4	24.1	27.2	31.2	35.9	34.0	32.6	31.6	28.3	23.5	18.0		
		MCWB	7.4	8.1	9.4	10.8	12.8	15.7	18.4	19.4	17.3	13.9	10.5	7.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	12.2	13.5	14.0	16.7	20.1	22.1	22.2	21.0	18.5	14.3	12.2		
		MCDB	17.1	20.0	25.8	28.8	31.7	30.4	29.0	30.1	28.4	25.6	21.9	17.8		
	2%	WB	10.5	10.9	11.9	12.9	15.2	19.0	21.4	21.6	20.3	17.4	13.1	10.7		
		MCDB	17.6	20.1	25.0	27.7	30.3	30.3	28.1	29.5	27.7	25.5	22.1	19.3		
	5%	WB	9.3	10.1	10.8	11.9	14.3	18.2	20.8	21.1	19.8	16.3	12.2	9.6		
		MCDB	17.9	19.9	23.5	26.6	30.2	30.5	27.5	28.9	27.1	25.3	22.0	18.5		
	10%	WB	8.3	9.0	10.1	11.1	13.3	17.3	20.4	20.6	19.1	15.2	11.3	8.6		
		MCDB	17.2	18.6	22.8	25.6	29.4	31.1	27.2	28.2	26.2	24.8	21.6	16.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	16.2	17.3	18.1	18.3	17.8	13.6	13.7	15.0	17.2	17.0	15.9		
		MCDBR	20.0	20.8	20.7	20.6	20.5	19.0	16.5	16.2	17.2	19.4	19.8	19.9		
		MCWBR	9.9	9.8	9.3	9.0	8.3	6.9	4.8	4.7	5.7	7.9	9.0	9.8		
	5% WB	MCDBR	16.0	16.9	18.5	18.5	18.7	15.5	12.4	13.3	13.3	14.9	16.0	16.4		
		MCWBR	8.4	8.2	8.3	8.2	7.9	5.3	3.5	4.1	4.5	6.0	7.5	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.259	0.278	0.287	0.302	0.331	0.386	0.373	0.329	0.293	0.265	0.253		
	taud		2.646	2.639	2.554	2.515	2.512	2.437	2.417	2.442	2.538	2.581	2.623	2.646		
	Ebn at Noon		996	1018	1010	1004	983	949	895	906	942	963	974	981		
	Edh at Noon		75	83	97	106	107	115	117	113	98	87	76	71		
(44) All-Sky Solar Radiation	RadAvg		3.74	4.67	6.16	7.52	8.25	8.07	6.51	6.31	5.87	5.14	4.11	3.43		
	RadStd		0.24	0.37	0.32	0.24	0.28	0.46	0.43	0.24	0.28	0.31	0.20	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.32	N/A	N/A	N/A	+0.52	+0.86	N/A	-161	+171	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-75	N/A	N/A

Nomenclature: See separate page